



Intelligent Control for DC Motor Systems

MCTA 4362 Machine Learning Project | Behavioral Cloning using Artificial Neural Networks |

2313703

Muhammad Nazhan

| Background and Problem Statement



Traditional Control

Classical PID controllers dominate industry due to simplicity but struggle with nonlinearities, aging components, and external disturbances.



Intelligent Solutions

Machine Learning provides an adaptive framework to learn complex system mappings from raw data without rigid mathematical constraints.

| Core Project Objectives

- ✓ **System Modeling:** Represent a high-performance DC motor using discrete-time state-space physics.
- ✓ **ML Framework:** Implement an ANN Behavioral Clone using supervised learning techniques.
- ✓ **Comparative Analysis:** Benchmark ML performance vs. classical PID under ideal and stress conditions.
- ✓ **GUI Terminal:** Develop a Gradio interface for real-time validation and stress testing.

DC Motor System Modeling

Plant Identification

A second-order transfer function maps input voltage $u(t)$ to angular velocity $\omega(t)$. Using Bilinear transformation for discretization:

$$G(s) = \frac{K}{(Js + b)(Ls + R) + K^2}$$

Resistance (R): $0.4 \, \Omega$

Inductance (L): $0.1 \, \text{H}$

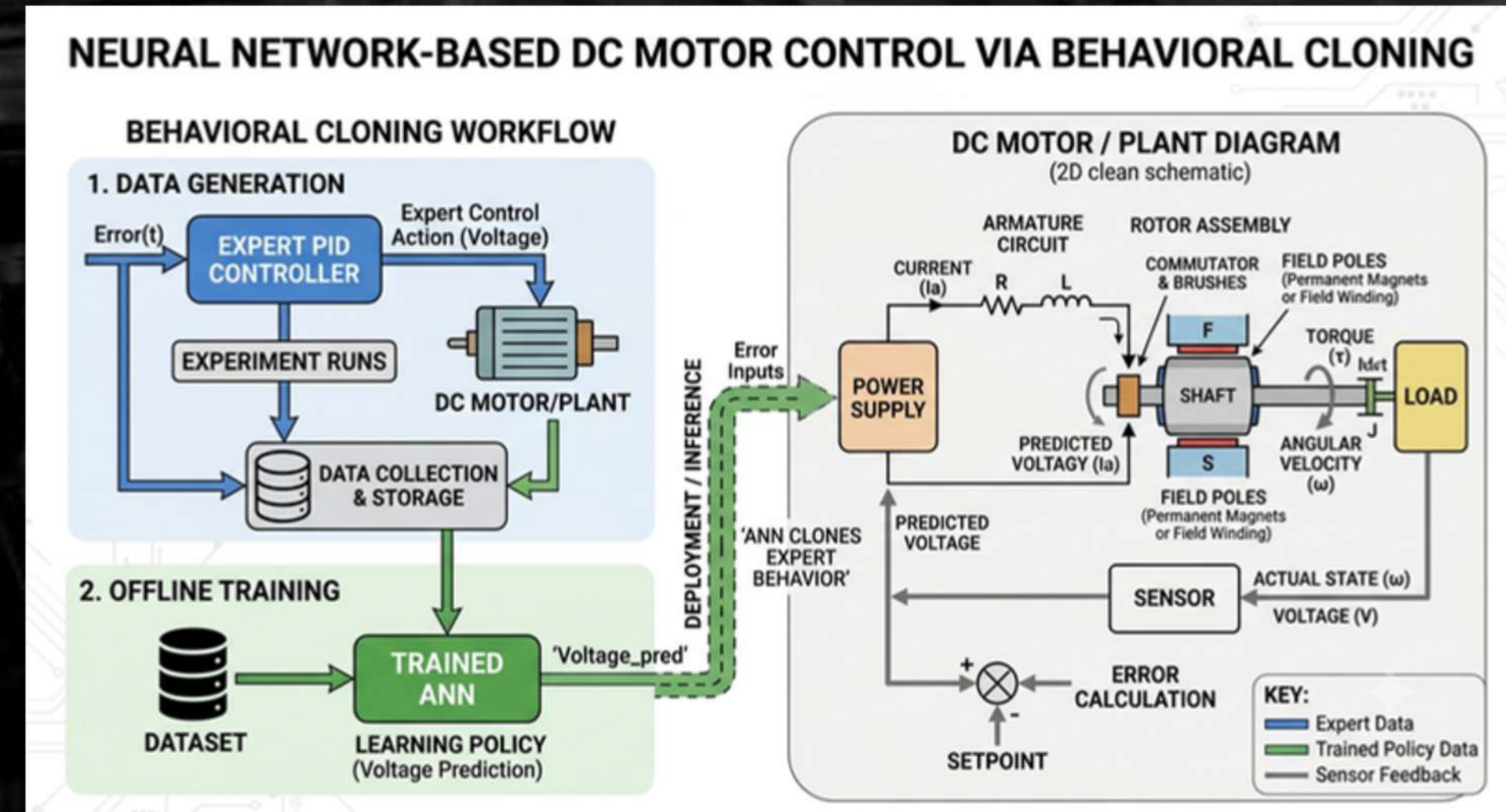
Inertia (J): $0.02 \, \text{kg} \cdot \text{m}^2$

Friction (b): $0.01 \, \text{N} \cdot \text{m} \cdot \text{s}$

Motor Constant (K): $0.15 \, \text{V/rad/s}$



$$G(s) = \frac{0.15}{0.002s^2 + 0.009s + 0.0265}$$



Classical PID Baseline Controller



Proportional

Responds to current error. Gain $K_p = 2.5$ ensures fast response but may cause overshoot.



Integral

Eliminates steady-state offset. Gain $K_i = 0.8$ handles accumulation of tracking error.



Derivative

Predicts future error trends. Gain $K_d = 0.1$ provides dampening against oscillations.

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

| Supervised ML: Behavioral Cloning

The Strategy

Instead of manual tuning, the AI learns to map the input/output policy of the PID directly.

Data Pipeline

Input Features: [Error, Integral, Derivative]

Target Labels: Optimal Voltage commands (u).

ANN Architecture & Training

Model Parameters

≡ **Layers:** 3-64-32-1 Topology

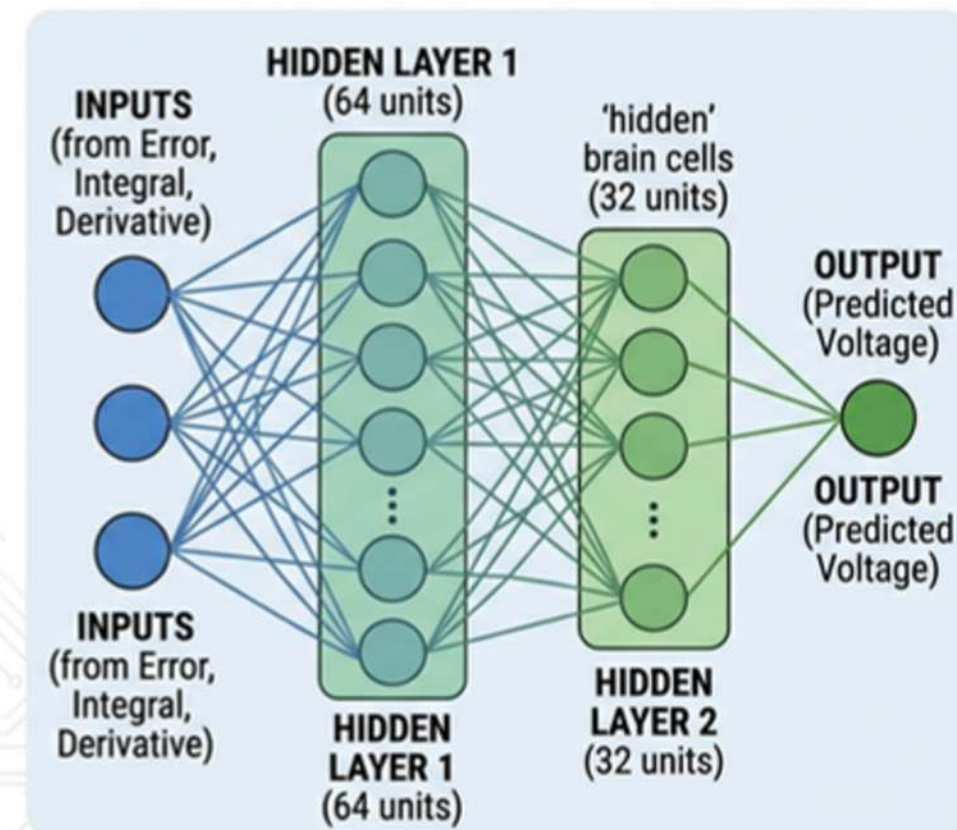
⚡ **Activation:** ReLU Functions

⚙️ **Solver:** Adam Optimizer

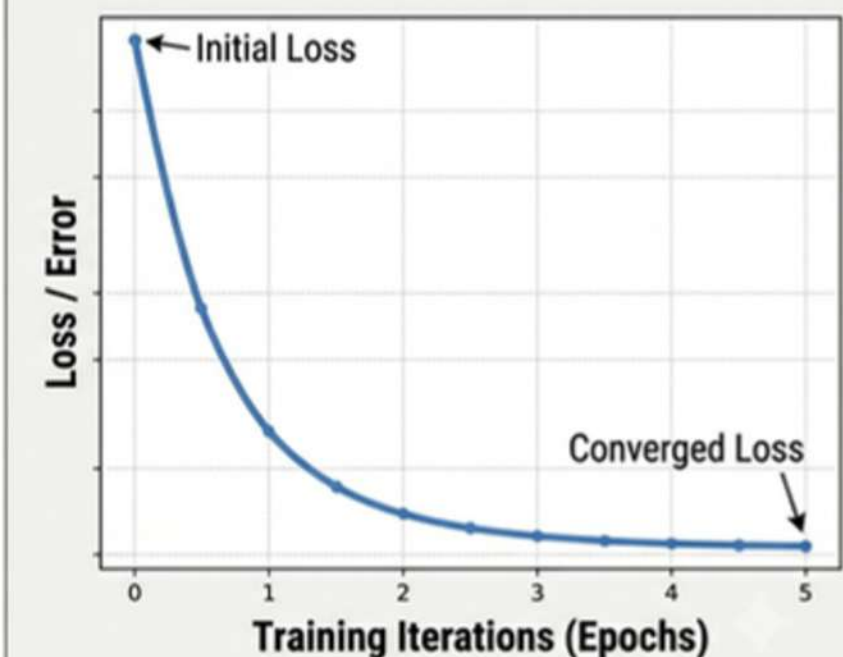
🗄️ **Training:** 3000 Iterations

NEURAL NETWORK-BASED DC MOTOR CONTROL VIA BEHAVIORAL CLONING

3-64-32-1 NETWORK ARCHITECTURE



TRAINING LOSS CURVE (flow converging and learning successfully)



| Stress Scenarios & Metrics

Evaluation Metrics

- Mean Squared Error (MSE)
- 🕒 Rise & Settling Time
- 📈 Peak Overshoot Percentage

Stress Scenarios

- ⚠️ **Ideal:** Noise-free tracking.
- ⚠️ **Noise:** Sensor interference.
- ⚠️ **Load:** Sudden torque disturbance.

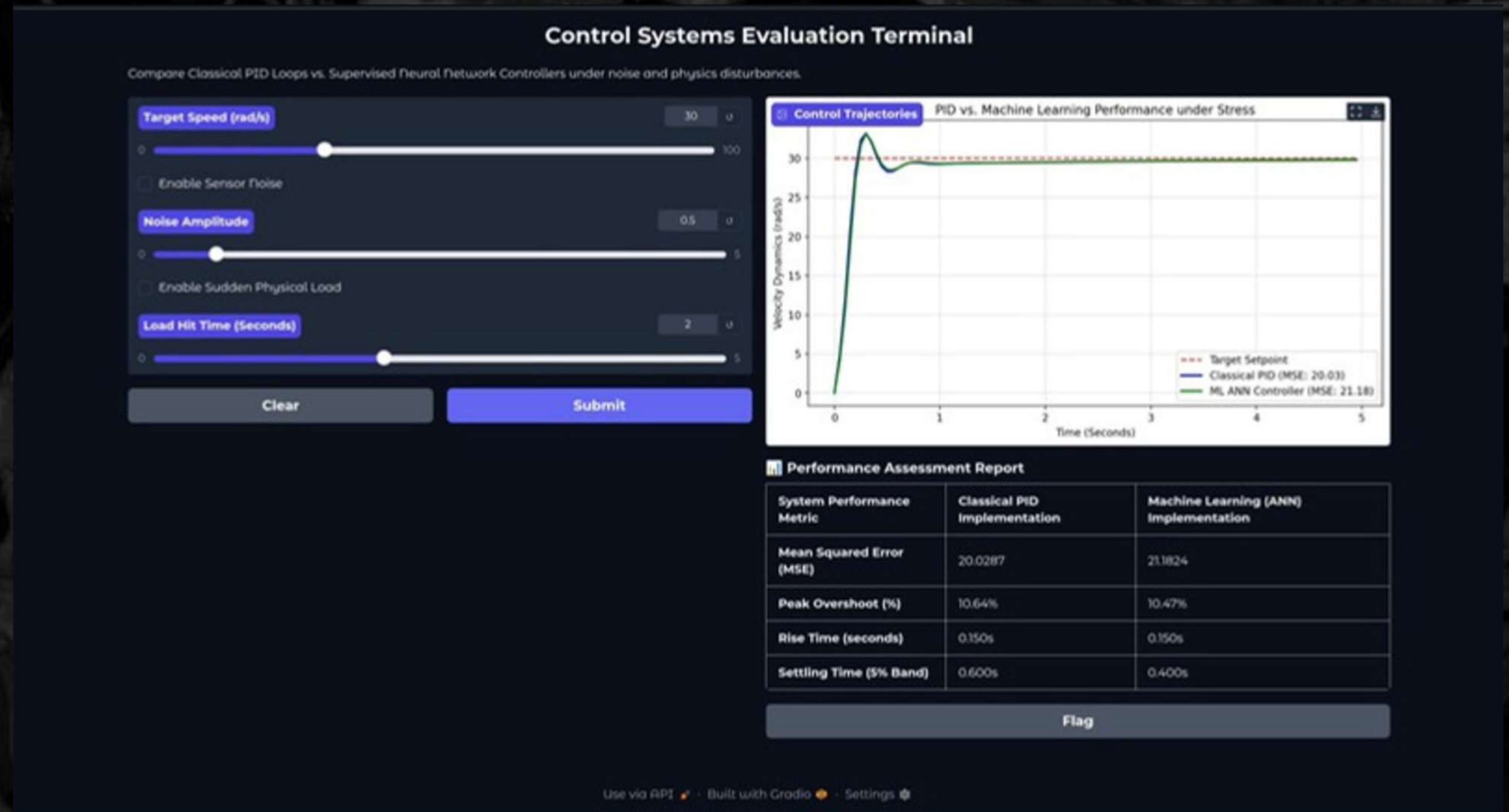
GUI USING GRADIO

INPUT

- Target Speed
- Noise Amplitude
- Load Hit Time

OUTPUT

- MSE
- Peak Overshoot
- Rise Time
- Settling Time



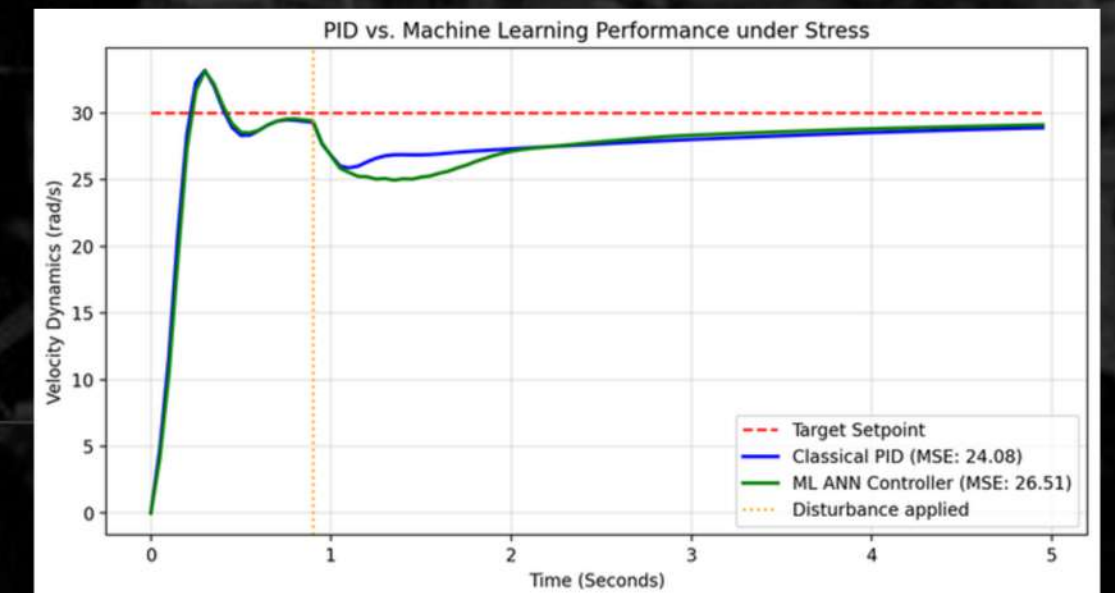
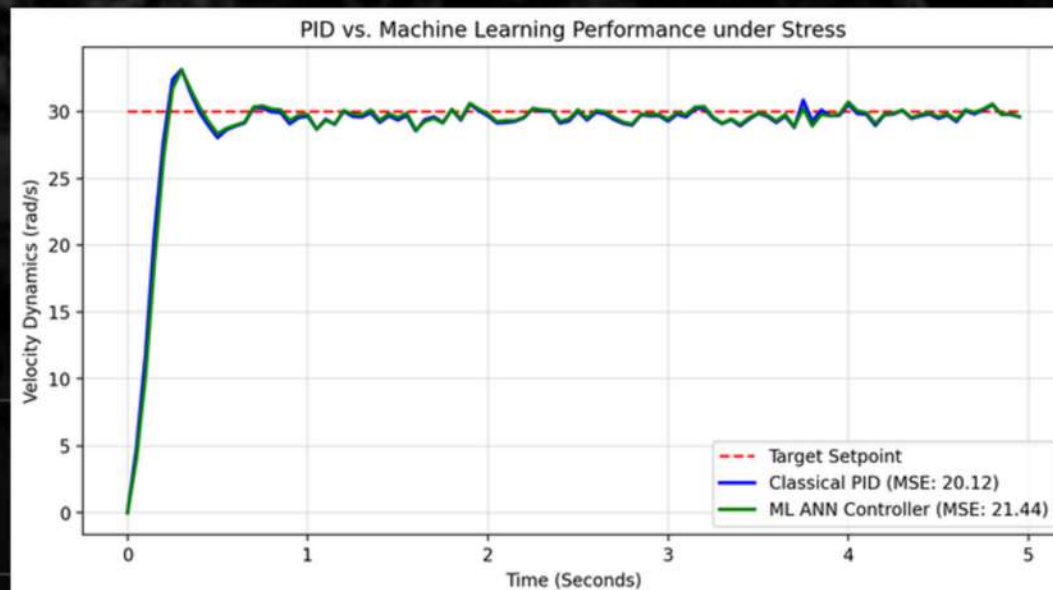
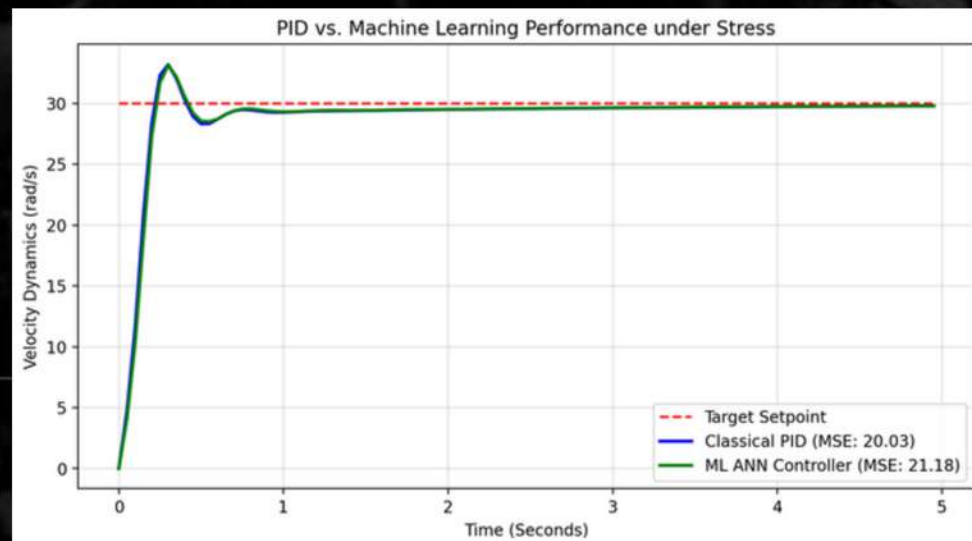
Comparative Performance Results

Clean Signal

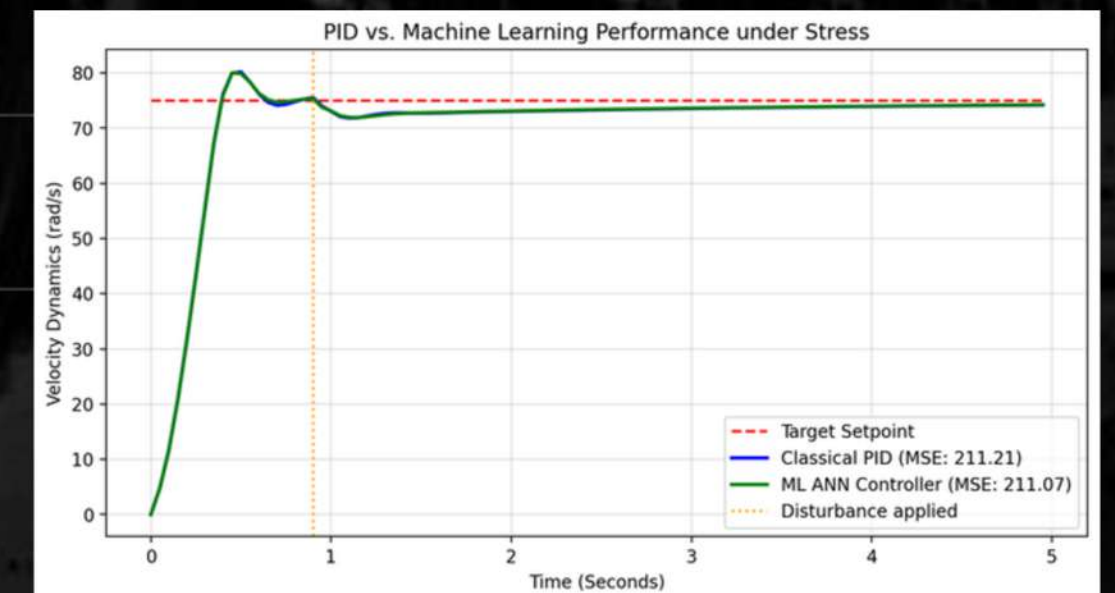
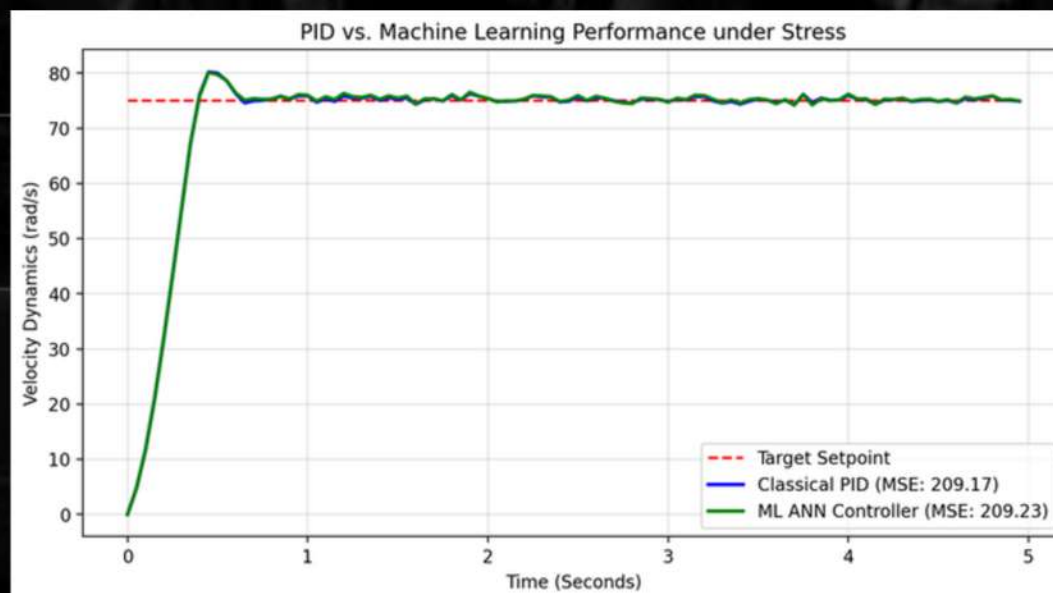
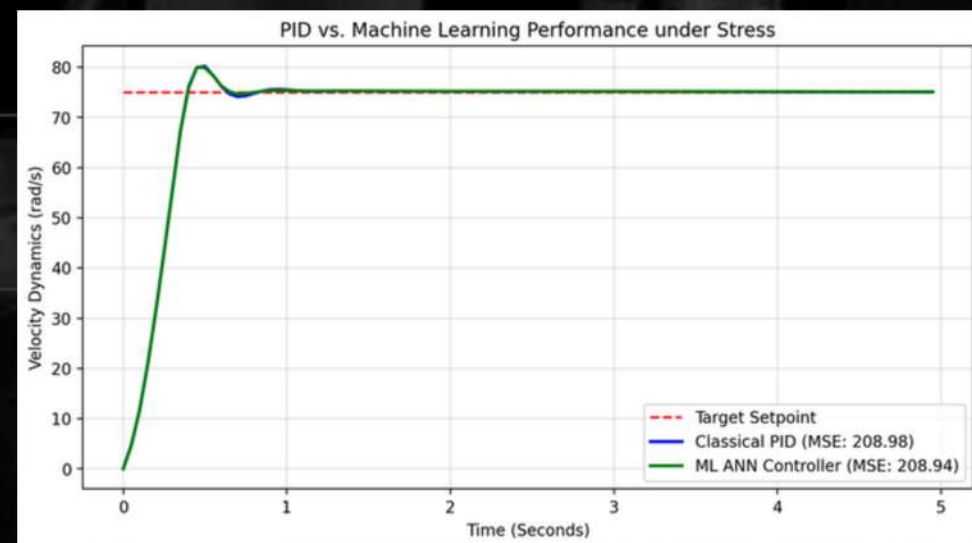
With Noise

With Load T=0.9

Target = 30



Target = 75



| Findings and Conclusions



Clone Success

ANN successfully cloned the PID policy, matching performance in ideal environments.



Robustness Gap

ML sensitivity to noise suggests a need for Data Augmentation or specialized filtering.



Future Work

Transition to Reinforcement Learning (DQN/PPO) to learn directly from environment interaction.

A dark, grayscale photograph of a car assembly line. Several car chassis are visible, moving along the line. The image is dimly lit, with the primary light source being the bright yellow text 'THANKYOU' centered in the middle. The background shows the industrial structure of the factory, including overhead lights and conveyor belts.

THANKYOU